

Programming Challenges

1. Retail Price Calculator

Build a JavaFX application where the user enters an item's wholesale cost and its markup percentage into two `TextField` controls. (For example, a \$5 wholesale cost with a 100% markup yields a \$10 retail price.) Add a `Button` that computes and displays the retail price when clicked.

2. Monthly Sales Tax

A retail company must file a monthly report listing total sales along with the state and county sales tax collected. The state rate is 4 percent and the county rate is 2 percent. Build a JavaFX application with a `TextField` for the month's total sales; when the user clicks a `Button`, display the county tax, the state tax, and the combined total tax. Represent the two rates as named constants in your code.

3. Property Tax

A county taxes property based on an assessed value equal to 60 percent of the property's market value — for example, a \$10,000 lot has an assessed value of \$6,000. The tax rate is \$0.64 per \$100 of assessed value, so that lot would owe \$38.40. Build a JavaFX application that asks the user for a property's market value and displays the assessed value and the tax owed.

4. Travel Expenses

Build a JavaFX application that calculates and displays a business traveler's total trip expenses. The user should supply:

- Number of days on the trip
- Airfare, if any
- Car rental fees, if any
- Miles driven, if a personal vehicle was used
- Parking fees, if any
- Taxi charges, if any
- Conference or seminar registration fees, if any
- Lodging charges, per night

The company reimburses according to this policy: \$47/day for meals; parking up to \$20/day; taxi charges up to \$40/day; lodging up to \$195/night; and \$0.42/mile if a personal vehicle was used. The application should display the total expenses incurred, the total allowable reimbursement, any excess the traveler must personally cover, and any savings if expenses came in under the allowance.

5. Theater Revenue

A movie theater keeps only a percentage of ticket revenue; the rest goes to the film distributor. Build a JavaFX application with TextField controls for the adult ticket price, the number of adult tickets sold, the child ticket price, and the number of child tickets sold. The application should compute and display the gross and net revenue for adult tickets, the gross and net revenue for child tickets, and the combined gross and net totals for the night. Assume the theater keeps 20 percent of box-office receipts, represented as a named constant.

6. Joe's Automotive

Joe's Automotive performs the following routine services: oil change — \$35.00; lube job — \$25.00; radiator flush — \$50.00; transmission flush — \$120.00; inspection — \$35.00; muffler replacement — \$200.00; tire rotation — \$20.00. Joe also performs nonroutine work billed for parts plus labor at \$60/hour. Build a JavaFX application that lets a customer select routine services and enter any nonroutine parts/labor charges, then displays the total for the visit.

7. Latin Translator

Given the word list below, build a JavaFX application that translates Latin to English. The GUI should have three Button controls, one per word; clicking one displays its English translation in a Label.

Latin	English
sinister	left
dexter	right
medium	center

8. MPG Calculator

Build a JavaFX application that calculates a car's fuel economy. The user enters the number of gallons the tank holds and the miles driven on that tank; clicking a *Calculate MPG* Button displays the miles-per-gallon figure, using $MPG = \text{Miles} \div \text{Gallons}$.

9. Name Formatter

Build a JavaFX application that collects a user's first name, middle name, last name, and preferred title (Mr., Mrs., Ms., Dr., etc.) into TextField controls. Given the sample data First name: *Kelly*, Middle name: *Jane*, Last name: *Smith*, Title: *Ms.*, the application should provide buttons that format and display the name as: "Ms. Kelly Jane Smith", "Kelly Jane Smith", "Kelly Smith", "Smith, Kelly Jane, Ms.", "Smith, Kelly Jane", and "Smith, Kelly".

10. Tip, Tax, and Total

Build a JavaFX application that lets the user enter a restaurant food charge into a TextField. When a Button is clicked, calculate and display an 18 percent tip, 7 percent sales tax, and the grand total of food charge, tip, and tax.

11. Distance Traveled

Assuming no delays, distance = speed $\times$ time. Build a JavaFX application that lets the user enter a car's speed in miles per hour, then provides buttons that display the distance traveled in 5 hours, 8 hours, and 12 hours.

12. Heads or Tails

Build a JavaFX application that simulates a coin toss. When the user clicks a Button, generate a random number in the range 0-1: 0 means heads, 1 means tails. Use an ImageView to display the corresponding coin face image.

13. Dice Simulator

Build a JavaFX application that simulates rolling a pair of dice. When the user clicks a Button, generate two random numbers in the range 1-6 and display each die's face using an ImageView (for example, Die1.png through Die6.png).



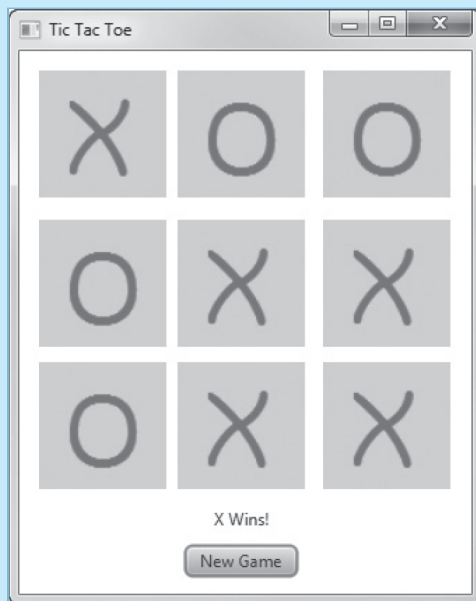
14. Celsius and Fahrenheit Converter

Build a JavaFX application that converts between Celsius and Fahrenheit. Provide a TextField for the temperature and two buttons: *Convert to Fahrenheit*, which treats

the entered value as Celsius and converts it using $F = 1.8 \times C + 32$, and *Convert to Celsius*, which treats the entered value as Fahrenheit and converts it using $C = (5/9) \times (F - 32)$.

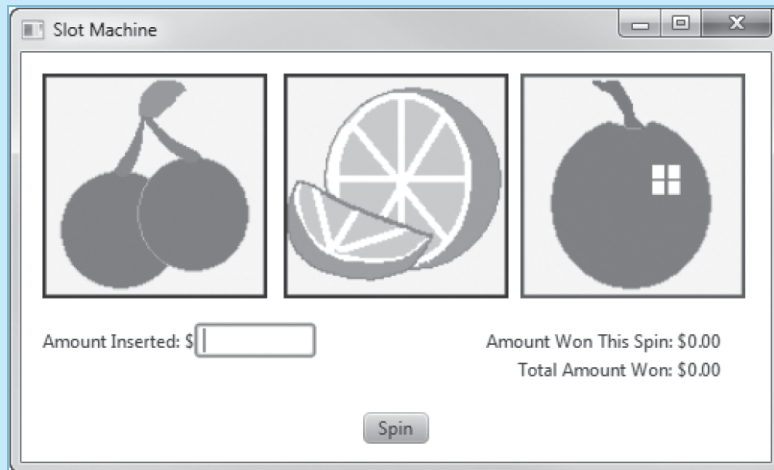
15. Tic-Tac-Toe Simulator

Build a JavaFX application that simulates a game of tic-tac-toe using nine `ImageView` controls arranged in a grid to display Xs and Os. Use a two-dimensional `int` array to hold the board in memory. When the user clicks *New Game*, fill the array with random 0s and 1s (0 = O, 1 = X), refresh the on-screen grid to match, and display a message announcing whether X won, O won, or the game was a tie.



16. Slot Machine Simulation

Build a JavaFX application that simulates a slot machine. The user enters the amount of money being inserted into a `TextField`. Clicking a *Spin* Button should randomly select three fruit symbols and display each in an `ImageView`. Award \$0 if no symbols match, twice the amount inserted if two symbols match, and three times the amount if all three match. Display both the amount won on the current spin and the running total won.



17. Vending Machine

Build a standalone JavaFX application that simulates a soft-drink vending machine stocking cola, lemon-lime soda, grape soda, root beer, and bottled water at \$0.75 each, with 20 units of each in stock at start-up. Provide a `TextField` for the amount of money inserted and a `Button` per drink. When a selection is made, dispense the drink (decrementing inventory) and show the change due in an `Alert` or `Label`. Handle operator errors gracefully — for example, selecting an out-of-stock drink, selecting with no money entered, or entering an insufficient amount.